

Estimating Gradients

A curve's gradient changes at every point: draw the tangent, make a big triangle, read rise over run.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Estimating Gradients

01 Gradient of a curve at a point

tangent at $x = 2$

zero gradient

equation of the tangent

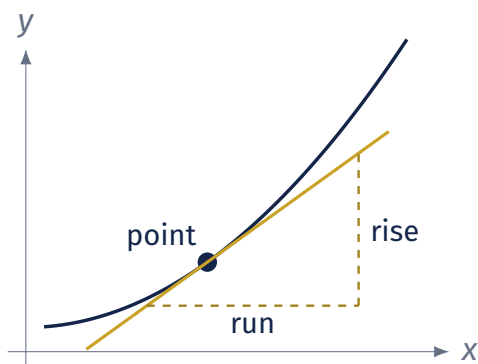
02 Rate of change in real life

speed from distance–time

acceleration from speed–time

interpret the gradient

THE TANGENT METHOD Three steps, one picture:



1. ruler touching the curve at the point
2. a BIG triangle on the tangent
3. gradient = $\frac{\text{rise}}{\text{run}}$ (with units)

ZONE distance--time

gradient = speed

m/s, km/h

ZONE speed--time

gradient = acceleration

m/s²

01 Gradient of a Curve at a Point

Lay a ruler so it touches the curve at the point only (tangent). Pick two far-apart points on the tangent and work out $\text{rise} \div \text{run}$. A going-down tangent has a negative gradient.

WHEN YOU SEE → YOU DO

“estimate the gradient of the curve at $x = 2$ ”

→ tangent, big triangle, $\text{rise} \div \text{run}$

“where is the gradient zero?”

→ the turning points (flat tangent)

“the gradient is negative”

→ between the maximum and the minimum

“find an equation of the tangent”

→ m from the triangle, then $y - y_1 = m(x - x_1)$

“a line through $(0, -17)$ is a tangent”

→ draw it from $(0, -17)$ just touching the curve

Draw the tangent

Big triangle

Rise $\div$ run

Sign check

WORKED MODEL • JANUARY 2020 P1HR

The curve passes through $P(2, 2.4)$. The tangent at P passes through $(0, 3.5)$ and $(6, 0.1)$.

$$1 \quad m = \frac{0.1 - 3.5}{6 - 0} = -\frac{3.4}{6}$$

rise $\div$ run

$$2 \quad m \approx -0.57; \quad c \approx 3.5$$

negative: going down

ANSWER $y \approx -0.6x + 3.5$

× **TRAP** Using a chord (a line through two points of the curve). The tangent touches at **one** point.

IDEA 01

Practise: Gradient of a Curve at a Point

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Tangent

CH 3 · TOPIC 6 3 marks

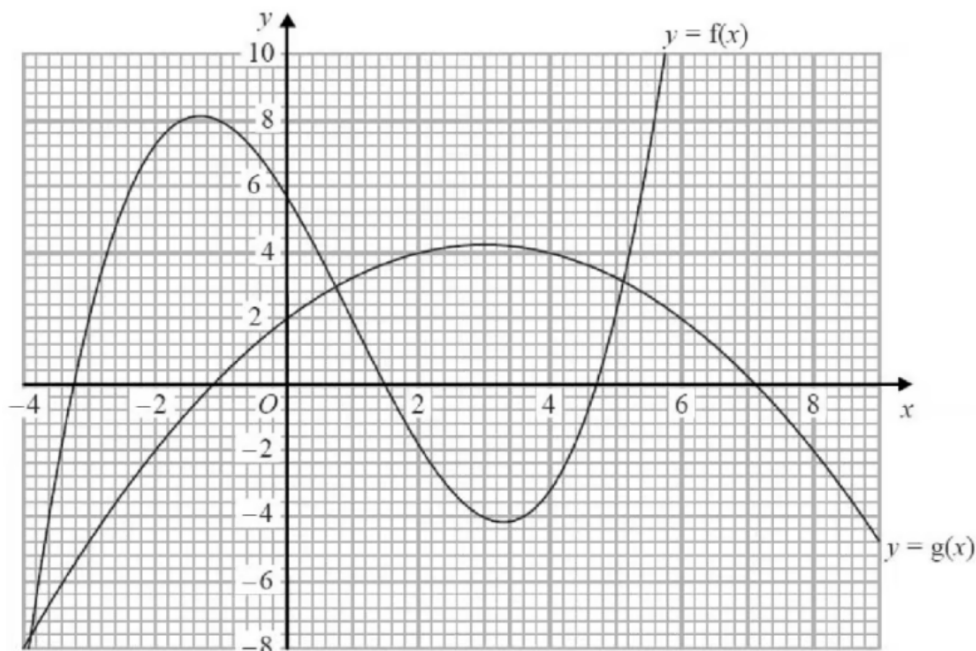
The tangent to a curve at $P(3, 5)$ also passes through $(0, -1)$. Find the gradient of the curve at P and the equation of the tangent.

SELECTED PRACTICE

Q2 PRACTICE Practice • Medium

CH 3 • TOPIC 6 3 marks

4 (a) The diagram shows parts of the graphs of $y=f(x)$ and $y=g(x)$



Write down the value of x where the gradient of the curve $y=g(x)$ is zero.

(1 mark)

(b) Calculate an estimate for the gradient of the curve $y = f(x)$ at the point on the curve where $x = 4$.

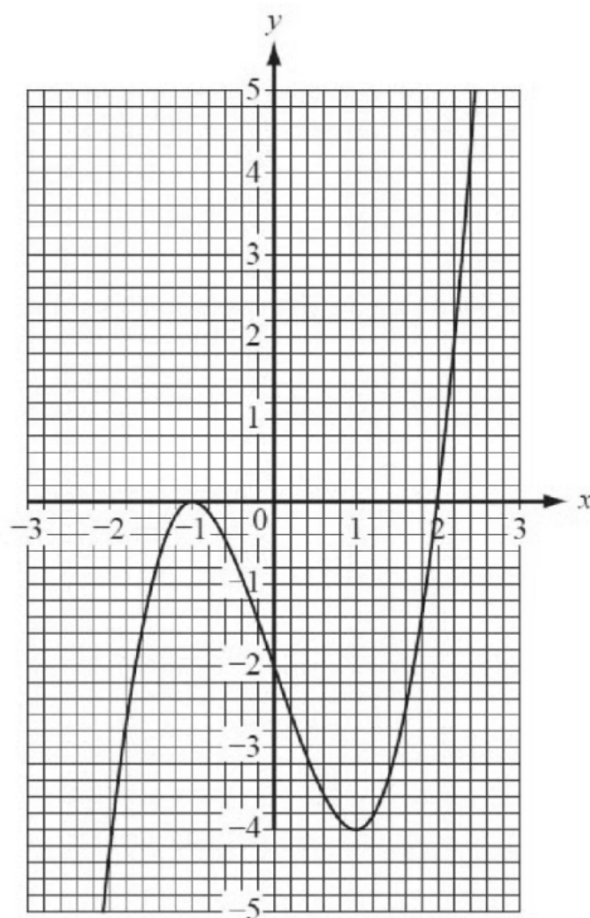
(2 marks)

Q3**PRACTICE**

Practice · Medium

CH 3 · TOPIC 6 5 marks

6 (a) The curve $y = x^3 - 3x - 2$ is shown on the grid.



Write down the co-ordinates of the points where the gradient of the curve is zero.

(2 marks)

(b) Write down the range of values of x when the gradient of the curve is negative.

(1 mark)

(c) Find an estimate of the gradient of the curve when $x = 2$.

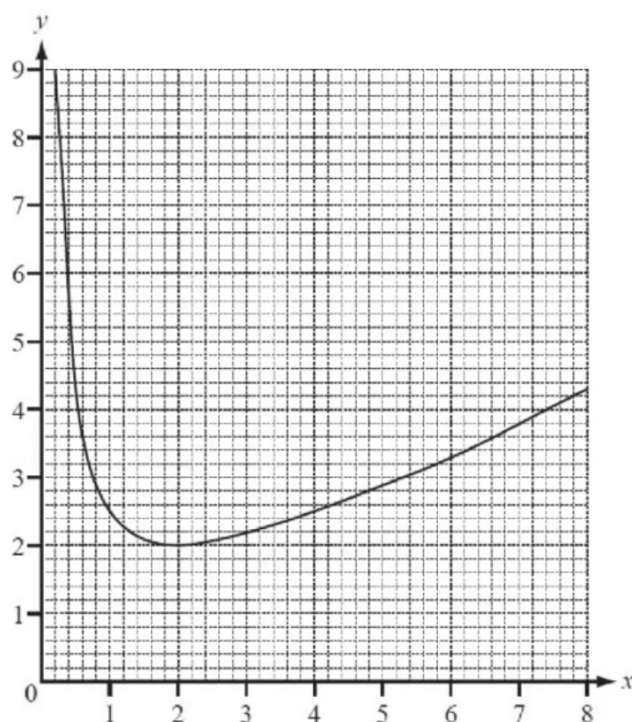
(2 marks)

Q4**PRACTICE**

Practice · Hard

CH 3 · TOPIC 6 5 marks

- 3 (a) The diagram shows the graph of $y = \frac{x}{2} + \frac{2}{x}$ for $0 < x \leq 8$.



Use the graph to solve the equation $\frac{x}{2} + \frac{2}{x} = 3$.

(2 marks)

- (b) By drawing a suitable tangent, find an estimate of the gradient of the graph when $x = 1$.

(3 marks)

Q5

PRACTICE

Practice • Very Hard

CH 3 • TOPIC 6

11 marks

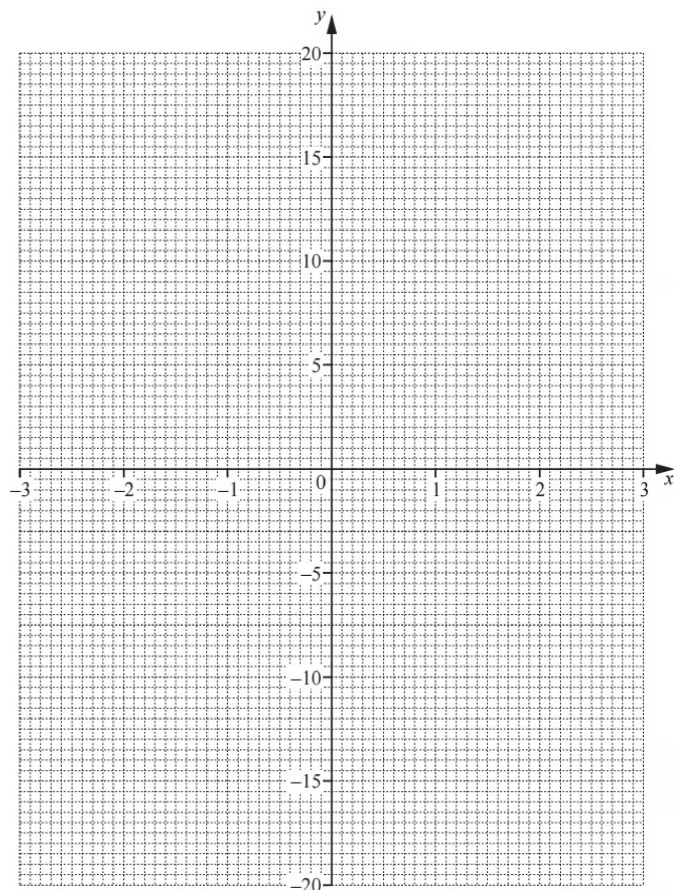
6 (a) The table shows some values of $y = x^3 - 3x - 1$.

x	-3	-2.5	-2	-1.5	-1	0	1	1.5	2	2.5	3
y	-19	-9.1		0.1	1	-1	-3	-2.1	1	7.1	

Complete the table of values.

(2 marks)

(b) Draw the graph of $y = x^3 - 3x - 1$ for $-3 \leq x \leq 3$.



(4 marks)

(c) A straight line through $(0, -17)$ is a tangent to the graph of $y = x^3 - 3x - 1$.

(i) On the grid, draw this tangent.

[1]

(ii) Find the co-ordinates of the point where the tangent meets your graph.

(.....,) [1]

(iii) Find the equation of the tangent. Give your answer in the form $y = mx + c$.

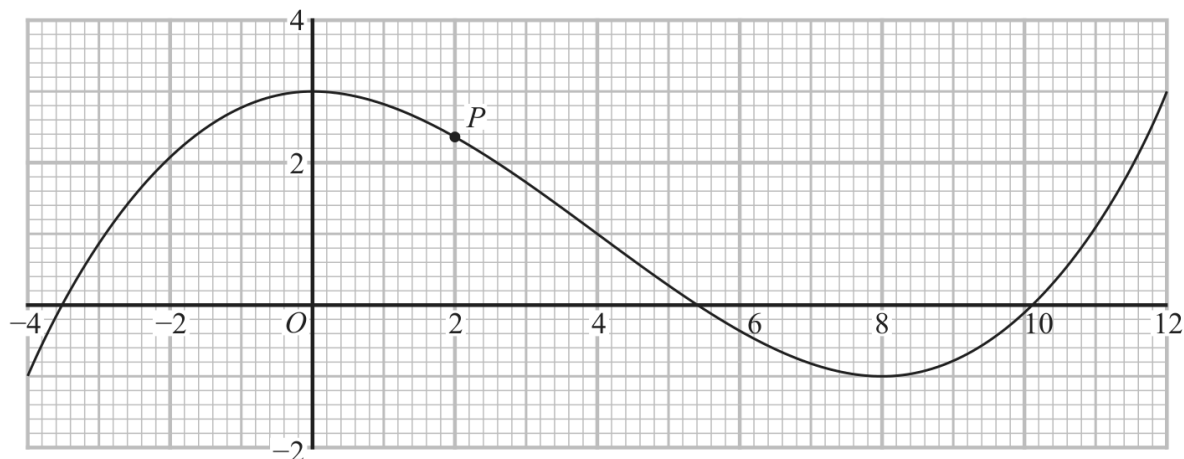
$y = \dots\dots\dots$ [3]

(5 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1HR • Q18

CH 3 • TOPIC 6 7 marks

18 The diagram shows the graph of $y = f(x)$ for $-4 \leq x \leq 12$ The point P on the curve has x coordinate 2(a) (i) Use the graph to find an estimate for the gradient of the curve at P .

.....

(ii) Hence find an equation of the tangent to the curve at P .Give your answer in the form $y = mx + c$

.....

The equation $f(x) = k$ has exactly two different solutions for $-4 \leq x \leq 12$ (b) Use the graph to find the two possible values of k .

..... ,

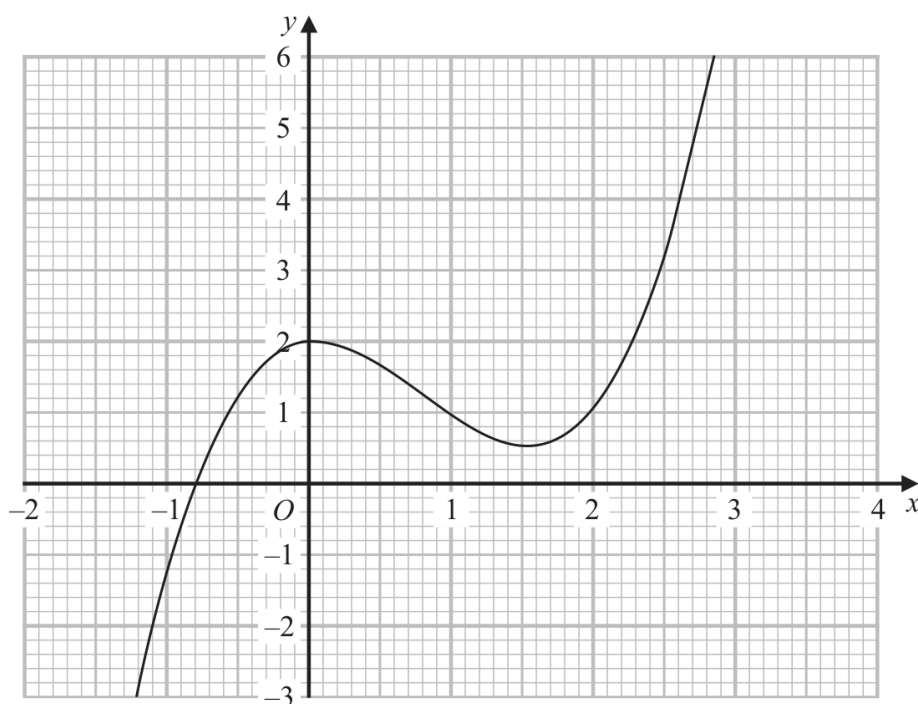
(Total for Question 18 is 7 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q17

CH 3 · TOPIC 6 3 marks

17 Part of the curve with equation $y = f(x)$ is shown on the grid.



Find an estimate for the gradient of the curve at the point where $x = 2$
Show your working clearly.

(Total for Question 17 is 3 marks)

02 Rate of Change in Real Life

The gradient has units: vertical units per horizontal unit. Always say what it means in the story.

WHEN YOU SEE → YOU DO

distance–time graph, “estimate the speed at 5 s”

→ gradient of the tangent (m/s)

speed–time graph, “the acceleration at $t = 9$ ”

→ gradient of the tangent (m/s^2); negative = slowing down

“interpret the gradient”

→ “the temperature rises by 3.5°C per hour at 3 hours”

“average rate between $t = 0$ and $t = 800$ ”

→ gradient of the chord between the two points

“why is it only an estimate?”

→ the tangent is drawn by eye

ZONE INSTANT

tangent at one point

“at $t = 3$ ”

ZONE AVERAGE

chord between two points

“between $t = 0$ and $t = 40$ ”

WORKED MODEL • AVERAGE ACCELERATION

A car’s velocity goes from 0 to 27 m/s in the first 40 seconds. Work out the average acceleration.

$$1 \quad \frac{27 - 0}{40 - 0}$$

chord: rise $\div$ run

ANSWER 0.675 m/s^2

IDEA 02

Practise: Rate of Change in Real Life

My questions first, then selected practice.

Q8

DR ESLAM Units

CH 3 • TOPIC 6 2 marks

On a distance–time graph (metres, seconds) the tangent at $t = 4$ has gradient 3.5. What does this mean?

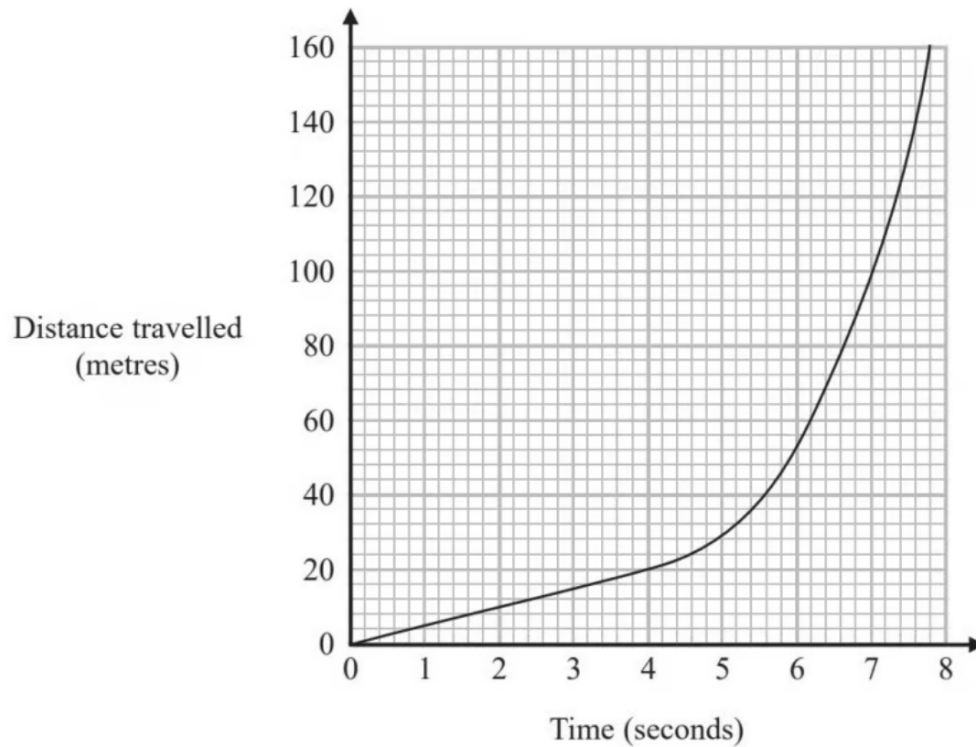
SELECTED PRACTICE

Q9

PRACTICE Practice · Medium

CH 3 · TOPIC 6 3 marks

- 1 The distance-time graph shows information about part of a car journey.



Use the graph to estimate the speed of the car at time 5 seconds.

(3 marks)

Q10

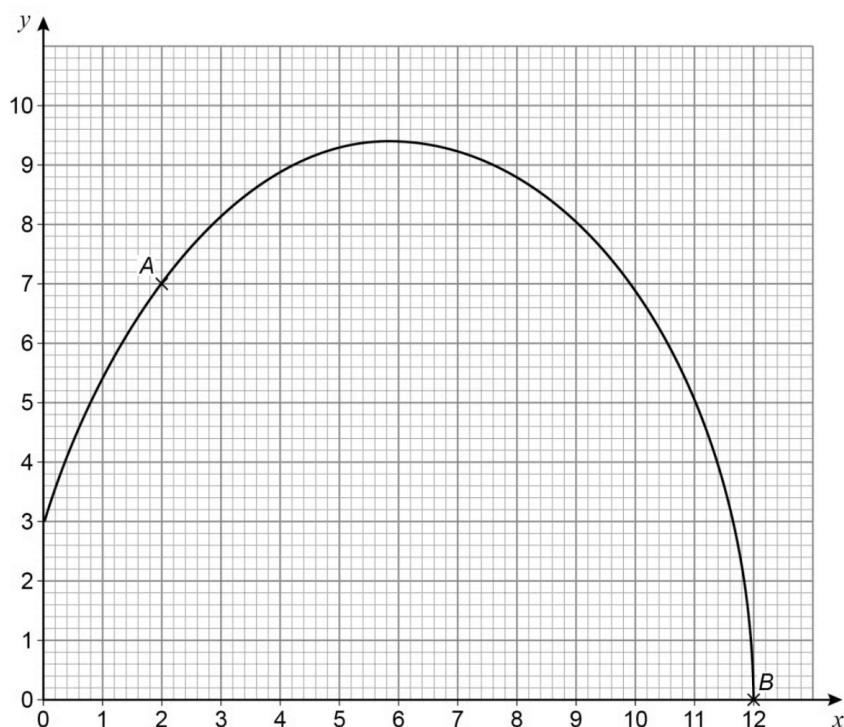
PRACTICE

Practice · Medium

CH 3 · TOPIC 6 3 marks

8 (a) A and B are points on a curve.

A is $(2, 7)$ B is $(12, 0)$



Work out the instantaneous rate of change of y with respect to x at point A .

(2 marks)

(b) The average rate of change of y with respect to x between points A and B is worked out.

Which statement is correct? Tick **one** box.

☐	It is positive.
☐	It is zero.
☐	It is negative.
☐	You cannot tell if it is positive or negative.

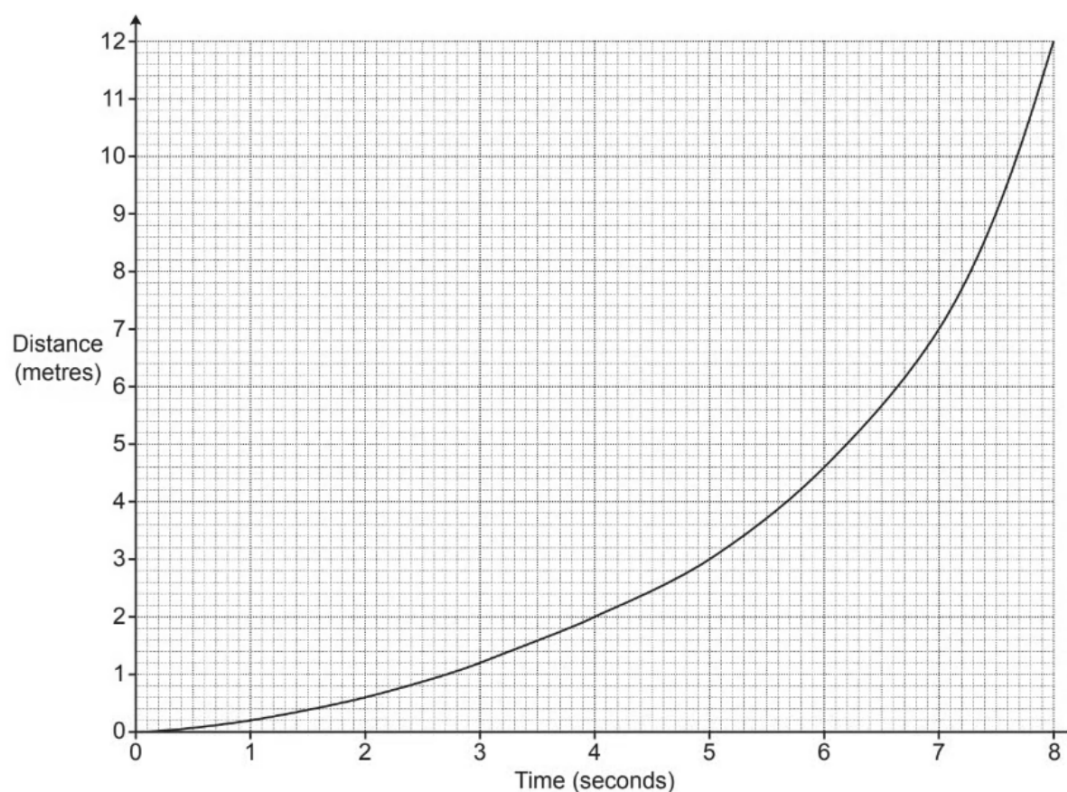
(1 mark)

Q11**PRACTICE**

Practice · Medium

CH 3 · TOPIC 6 4 marks

11 The graph shows the distance travelled by a particle over 8 seconds.



Estimate the speed of the particle at 5 seconds.

..... m/s
(4 marks)

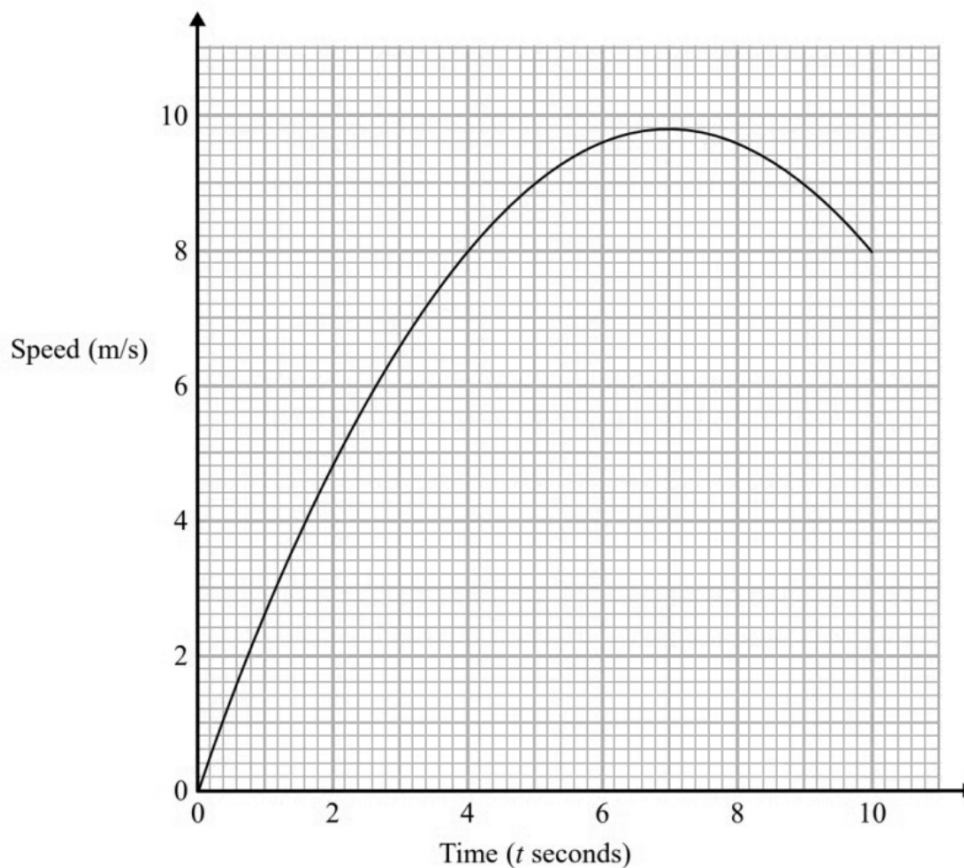
Q12**PRACTICE**

Practice · Hard

CH 3 · TOPIC 6 6 marks

4 (a) Karol runs in a race.

The graph shows her speed, in metres per second, t seconds after the start of the race.



Calculate an estimate for the gradient of the graph when $t = 4$. You must show how you get your answer.

(3 marks)

(b) Describe fully what your answer to part (a) represents.

(2 marks)

(c) Explain why your answer to part (a) is only an estimate.

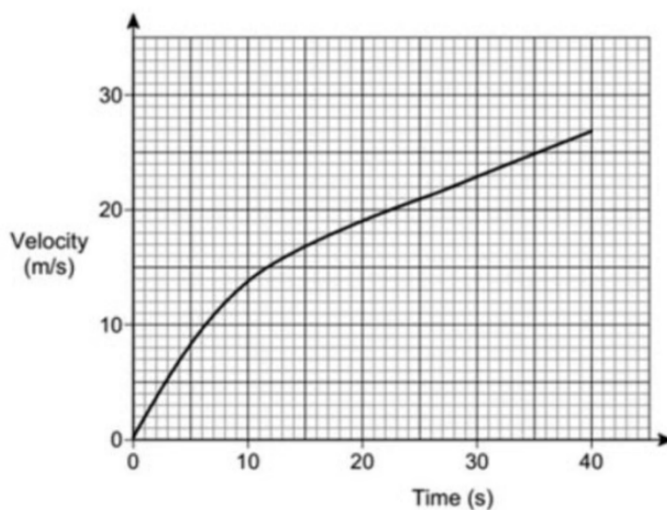
(1 mark)

Q13**PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 6 5 marks

- 1 (a) The velocity-time graph shows the first 40 seconds of a car in a race.



Work out the average acceleration for the first 40 seconds.
Give the units of your answer.

(3 marks)

- (b) Estimate the time during the 40 seconds when the instantaneous acceleration = the average acceleration.

You must show your working on the graph.

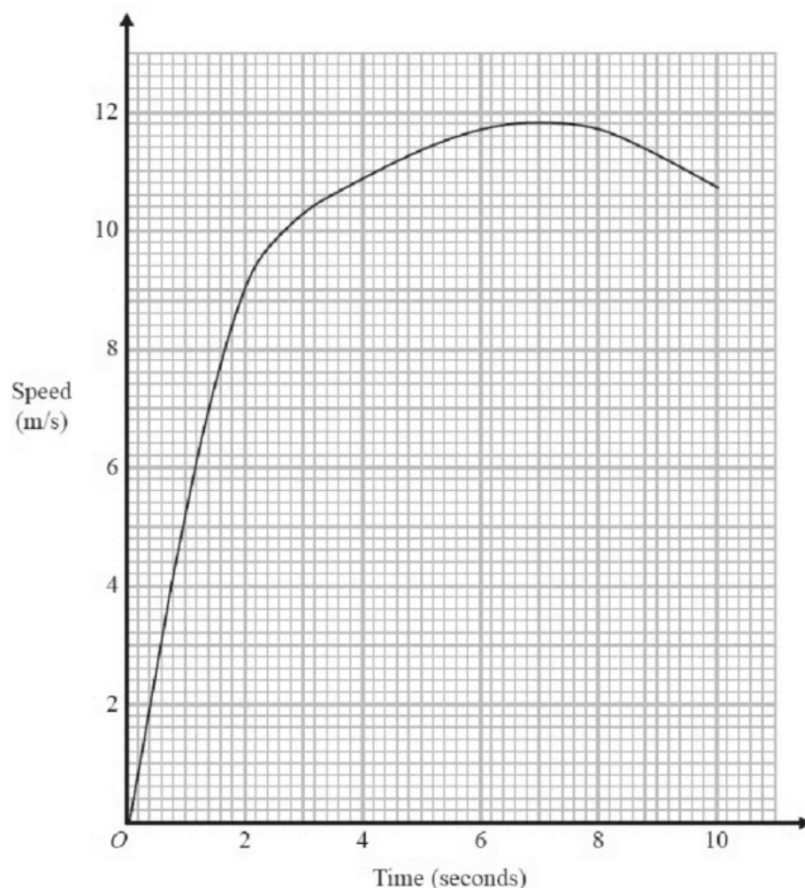
(2 marks)

Q14 PRACTICE Practice • Very Hard

CH 3 • TOPIC 6 7 marks

5 (a) Olympic medallist Usain runs in a race.

The graph shows his speed, in metres per second (m/s), during the first 10 seconds of the race.



Use the graph to find how long it took Usain to reach his top speed.

(1 mark)

(b) Work out an estimate for Usain's acceleration at 2 seconds. Give the units of your answer.

(3 marks)

(c) Calculate the difference in Usain's acceleration between 2 and 6 seconds.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

01 At a point

Tangent, big triangle, rise $\div$ run; zero at turning points.

02 Real life

Units and meaning; tangent = instant, chord = average.

BOTTOM LINE Tangent $\rightarrow$ big triangle $\rightarrow$ rise $\div$ run $\rightarrow$ units $\rightarrow$ what it means.

Answers

Estimating Gradients · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • GRADIENT OF A CURVE AT A POINT

- Q1** Dr Eslam · Tangent $m = \frac{5+1}{3} = 2; y = 2x - 1$
- Q2** Practice · Medium (a) $x \approx 3$ (b) about 5 (tangent estimate)
- Q3** Practice · Medium (a) $(-1, 0)$ and $(1, -4)$ (b) $-1 < x < 1$ (c) 9
- Q4** Practice · Hard (a) $x \approx 0.8, 5.2$ (b) -1.5
- Q5** Practice · Very Hard (a) $-3, 17$ (c)(ii) $(2, 1)$ (iii) $y = 9x - 17$
- Q6** Edexcel 4MA1 · January 2020 · Paper 1HR · Q18 (a)(i) about -0.6 (ii) $y \approx -0.6x + 3.5$ (b) $k = 3$ and $k = -1$
- Q7** Edexcel 4MA1 · January 2022 · Paper 2HR · Q17 about 2.7 (tangent estimate)

IDEA 02 • RATE OF CHANGE IN REAL LIFE

- Q8** Dr Eslam · Units at 4 seconds the speed is 3.5 m/s
- Q9** Practice · Medium about 15 m/s (tangent at 5 s)
- Q10** Practice · Medium (a) about 1.3 (b) negative
- Q11** Practice · Medium about 1.3 m/s
- Q12** Practice · Hard (a) about 1.3 (b) her acceleration at 4 s, in m/s^2 (c) the tangent is drawn by eye
- Q13** Practice · Very Hard (a) $\frac{27}{40} = 0.675 \text{ m/s}^2$ (b) about 12 s (tangent parallel to the chord)
- Q14** Practice · Very Hard (a) about 7 s (b) about 1.5 m/s^2 (c) about 1.2 m/s^2